Increment 1 – Test Report

Status: Complete · Oct 24, 2026 · Author: Salam Alsaour
Input: 01_Requirements.md (acceptance tests, Section 6; demo scenario, Section 7; Definition of Done, Section 8), 02_Design.md (test design, Section 11)
Build tested: branch `increment-1` ; backend Python 3.11, PostgreSQL 16; frontend Node 22, React 18, Vite 8

1. Summary

Check	Result
Backend tests (pytest, real PostgreSQL)	**108 passed**, 0 failed
Backend line coverage	**95%** of `app` (gate 70%; seed, CLI, scheduler wiring, and app factory excluded by configuration)
Permission matrix	107 routes listed; **494** calls by callers that are not allowed, all answered `401` or `403`
Acceptance tests of 01_Requirements Section 6	**43 of 43 passed** (automated, plus the browser checks of AT-AUT-01, AT-PRF-05, and AT-LOC-01)
Non-functional checks of 01_Requirements Section 4.2	14 of 14 met; NFR-MNT-01 is confirmed when CI runs on the pull request
Frontend	TypeScript strict build clean; **7 Vitest tests passed** (translation keys and date format); production build succeeds
Lint	`ruff` clean on `app` , `tests` , `alembic` , `scripts`
M3 demo dry run on `docker compose up`	**10 of 10 steps passed**, twice in a row, with real emails through Mailpit
Dependencies shipped to users (`npm audit --omit=dev`)	0 known vulnerabilities
Committed scope	39 Must requirements delivered; 4 of 4 stretch items delivered (FR-ORG-09, FR-CYC-05, FR-VAC-05, FR-VAC-08); FR-ORG-07 deferred to Increment 3 as planned

2. How to Repeat the Tests

```
docker compose up -d db mailpit          # PostgreSQL on 5433, Mailpit on 8025
cd backend
pip install -r requirements-dev.txt
ruff check app tests alembic scripts
pytest --cov=app                         # creates the internsync_test database on first use
cd ../frontend
npm ci
npx tsc -b && npx vitest run && npx vite build
```

The M3 rehearsal against the full stack:

```
docker compose up -d --build
docker compose exec api python -m app.seed --demo
python backend/scripts/demo_dry_run.py   # needs httpx (in requirements-dev.txt)
```

3. Acceptance Tests

Every test of 01_Requirements Section 6 has a named test in `backend/tests` (the AT ID is in the function name). "Browser" means the check was also done by hand in the running app, in Arabic and English.

Test	Automated test(s)	Result
AT-AUT-01	`test_at_aut_01_each_role_logs_in_with_its_role` ; browser: each role's menu, the team page only for the Company Admin	Pass
AT-AUT-02	`test_at_aut_02_registration_kinds_and_single_use_confirmation` , `test_at_aut_02_expired_confirmation_link` , `test_resend_confirmation`	Pass
AT-AUT-03	`test_at_aut_03_idle_timeout_and_logout` , `test_session_max_lifetime`	Pass
AT-AUT-04	`test_at_aut_04_password_reset` , `test_at_aut_04_expired_reset_link`	Pass
AT-AUT-05	`test_every_route_is_in_the_matrix_and_nothing_else` , `test_at_aut_05_disallowed_callers_are_refused` , `test_allowed_callers_pass_the_role_gates` ; scope cases in the module suites	Pass
AT-AUT-06	`test_at_aut_06_deactivation` , `test_deactivated_user_session_ends` , `test_company_suspend_and_reactivate`	Pass
AT-ORG-01	`test_at_org_01_new_college_appears_and_deactivated_does_not` , `test_registration_refused_for_inactive_college`	Pass
AT-ORG-02	`test_at_org_02_unverified_accounts_are_limited`	Pass
AT-ORG-03	`test_at_org_03_one_coordinator_per_college`	Pass
AT-ORG-04	`test_at_org_04_company_verification`	Pass
AT-ORG-05	`test_at_org_05_student_verification_enrols` , `test_student_request_waits_without_open_cycle`	Pass
AT-ORG-06	`test_at_org_06_invited_supervisor_joins_the_college` , `test_at_org_06_expired_invitation_and_resend` , `test_company_admin_invites_colleague`	Pass
AT-ORG-08	`test_at_org_08_document_access_and_purge` (clock moved 29 and 31 days)	Pass
AT-ORG-09	`test_at_org_09_resubmission_after_rejection`	Pass
AT-CYC-01	`test_at_cyc_01_unit_weeks_follow_weekday_and_time` , `test_at_cyc_01_api_creates_twelve_weeks` , `test_unit_partial_last_week_counts_only_days_in_the_cycle`	Pass
AT-CYC-02	`test_at_cyc_02_weights_must_total_100`	Pass
AT-CYC-03	`test_at_cyc_03_locked_rubric_cannot_change`	Pass
AT-CYC-04	`test_at_cyc_04_unit_synonym_matches_and_shows_canonical_name` , `test_skill_catalog_admin`	Pass
AT-CYC-05	`test_at_cyc_05_closed_cycle_is_read_only` , `test_closed_cycle_freezes_reports`	Pass
AT-PRF-02	`test_at_prf_02_verified_student_cannot_edit_academic_fields` , `test_pending_student_can_correct_academic_fields`	Pass

Test	Automated test(s)	Result
AT-PRF-03	`test_at_prf_03_size_and_type_limits_and_versions`	Pass
AT-PRF-04	`test_at_prf_04_unit_full_dictionary_on_sample_cvs`, `test_at_prf_04_extracted_skills_are_only_candidates`, `test_arabic_docx_cv_extraction`	Pass
AT-PRF-05	`test_at_prf_05_image_only_pdf`; browser: the message and the empty editable list	Pass
AT-VAC-01	`test_at_vac_01_create_and_validate`	Pass
AT-VAC-02	`test_at_vac_02_visibility`	Pass
AT-VAC-03	`test_at_vac_03_filters`	Pass
AT-VAC-04	`test_at_vac_04_application_rules` (duplicate, wrong training type, after deadline, sixth application), `test_unit_eligibility_reasons`	Pass
AT-VAC-05	`test_at_vac_05_withdraw_frees_a_slot`	Pass
AT-VAC-06	`test_at_vac_06_applicants_and_decisions`	Pass
AT-VAC-07	`test_at_vac_07_placement_rules`, `test_placement_with_another_company_supervisor`	Pass
AT-VAC-08	`test_at_vac_08_change_supervisor_and_cancel`	Pass
AT-REP-01	`test_at_rep_01_slots_and_deadlines`	Pass
AT-REP-02	`test_at_rep_02_attachment_limits`, `test_weekly_report_needs_all_fields`, `test_final_report_needs_a_main_document`	Pass
AT-REP-03	`test_at_rep_03_drafts_private_and_submission_final`	Pass
AT-REP-04	`test_at_rep_04_unit_deadline_minute_is_inclusive`, `test_at_rep_04_on_time_late_and_missing`	Pass
AT-REP-05	`test_at_rep_05_revision_cycle_keeps_versions`	Pass
AT-REP-06	`test_at_rep_06_comments`	Pass
AT-REP-07	`test_at_rep_07_closed_slot_refuses_submission`	Pass
AT-WFL-01	`test_at_wfl_01_every_allowed_transition_passes`, `test_at_wfl_01_disallowed_transitions_from_each_state_are_rejected`	Pass
AT-LOC-01	`test_language_is_saved_and_used_for_errors`; browser: every Increment 1 page in both languages, right to left in Arabic and left to right in English	Pass

Test	Automated test(s)	Result
AT-LOC-02	`frontend/src/i18n/keys.test.ts` (same keys in both files, every key used in the code exists, placeholders match); 506 keys	Pass
AT-AUD-01	`test_m3_demo_scenario_and_at_aud_01` ; dry-run step 10	Pass
AT-AUD-02	`test_at_aud_02_no_endpoint_changes_audit_entries` , `test_at_aud_02_database_refuses_update_and_delete` , `test_at_aud_02_filters_and_scope`	Pass

4. Non-Functional Checks

ID	Evidence	Result
NFR-SEC-02	Argon2id hashing (`app/core/security.py`); `test_weak_password_refused`	Met
NFR-SEC-03	`test_lockout_after_five_failures` (locked, then open again after 16 minutes); registration limit `test_registration_rate_limit`	Met
NFR-SEC-04	Type detected from content (`test_at_prf_03…` refuses a PNG named `.pdf` , a ZIP named `.docx` , and text); files stored under random keys outside the web root; `403` without permission (`test_at_org_08…` , `test_cv_file_visible_to_company_only_after_application`); downloads send `nosniff`	Met
NFR-PRV-01	`test_consent_gate` ; privacy notice page at first login	Met
NFR-PRV-02	Text extraction and skill matching run inside the API (`app/ai`); no outside service is called	Met
NFR-PRV-03	Permission matrix; the Platform Admin is refused on CV, report, and placement routes; audit entries of the college only for its coordinator (`test_at_aud_02_filters_and_scope`)	Met
NFR-REL-04	A refused attachment (too large or wrong type) leaves the draft and its other files unchanged (`test_at_rep_02_attachment_limits`)	Met
NFR-L10N-01	Browser check of every page in both languages and directions	Met
NFR-L10N-02	Deadlines stored in UTC and shown in Asia/Riyadh as DD/MM/YYYY HH:mm (`format.test.ts`); 1 minute before is on time, 1 minute after is late (`test_at_rep_04_unit_deadline_minute_is_inclusive`)	Met
NFR-MNT-01	`.github/workflows/ci.yml` runs lint, tests with the 70% gate, and the frontend checks on every pull request; local coverage 95%	Met locally; CI run on the pull request
NFR-MNT-02	OpenAPI generated by FastAPI; the frontend types are generated from it, and CI fails if either committed copy is out of date	Met
NFR-MNT-03	Link validity, session timeout, lockout, application limit, file sizes, attachment count, and purge days are read from `system_settings` (`test_majors_and_settings` , `test_at_vac_05…` changes the limit)	Met
NFR-PRT-02	`docker compose up -d --build` starts database, API (migrations and reference data on start), web app, and Mailpit; the dry run ran against it	Met
NFR-USA-03	Checked at 375 px (phone, the sidebar stacks above the content) and 1280 px	Met

5. M3 Demo Dry Run

Run with `backend/scripts/demo_dry_run.py` against `docker compose up` on Oct 24, 2026. Emails were read from Mailpit, and the demo clock was moved past the week 2 deadline for step 8.

Step	Result
1. Platform Admin adds a college and a skill	Pass
2. Coordinator registers with an appointment letter, confirms the email, is approved	Pass
3. 12-week Co-op cycle with 12 deadlines, weights 40/30/30, a rubric; invited academic supervisor joins	Pass
4. Company registers and is approved; Company Admin invites a colleague; vacancy published for the university	Pass
5. Student registers, sees the pending status, completes the profile, uploads a CV (6 skills suggested), confirms the skills	Pass
6. Coordinator approves and enrols the student; the filtered list shows the vacancy; the student applies	Pass
7. Company accepts; coordinator confirms the placement; 13 report slots appear	Pass
8. Week 1 submitted late, revision requested, resubmitted (2 versions kept), accepted; week 2 closed as Missing	Pass
9. Language switched to English and back, saved on the account	Pass
10. Coordinator's audit log shows the student's 14 entries covering every step	Pass

The same steps were also done by hand in the browser for the student (writing and submitting a report) and the academic supervisor (requesting a revision), and every page of the coordinator, company, and Platform Admin was opened in both languages.

6. Defects Found and Fixed

Defect	Found by	Fix
Sessions ended at once when the server's clock and the application clock differed (token issue time checked by the JWT library)	`test_at_aut_01…`	Expiry checked against the application clock only
Error messages ignored the user's saved language	`test_language_is_saved_and_used_for_errors`	Language kept on the request and applied when the error is written
Publishing a vacancy in the request that created it failed	Demo seed	Database session flushes pending rows before queries
A failed job could leave its lock held on a pooled connection, so the job never ran again	`test_job_failure_is_recorded_and_releases_the_lock`	Transaction-level advisory lock
The API container stopped at start with a plain `CORS_ORIGINS` value	`docker compose up`	Comma-separated values accepted; `test_config.py`
The Missing flag waited 15 minutes after every restart	Dry run on Docker	First run 30 seconds after start
Fractions such as `3.10 / 5.00` were reversed on Arabic pages	Browser check	Shown left to right
Single-word synonyms such as "rest", "spring", and "العربية" matched ordinary CV text	Review of the skill vocabulary	Removed; "Go" and "R" renamed

7. Known Limitations and Follow-Up

- **Development tooling advisories:** `npm audit` reports 3 moderate advisories in Tailwind CSS 3 and its PostCSS helpers, which run only while the site is built. The fix is the move to Tailwind 4, planned for Increment 2. Vite and Vitest were upgraded in this increment for the same reason.
- **Backend type checking:** the frontend is type-checked in strict mode; the backend relies on `ruff` and the tests. A type checker for the backend is planned for Increment 2.
- **Browser tests:** the end-to-end checks of this increment were done by hand. Automated browser tests are planned with the Increment 2 screens.
- **Skill extraction accuracy** is measured at M4 on labelled CVs (decision RQ-4).
- **Performance and HTTPS** are checked on the free-tier deployment at the start of Increment 2 (decision RQ-7).
- **Email delivery** is tested through Mailpit; the real SMTP account is configured with the deployment.

8. Definition of Done

Item (01_Requirements Section 8)	Status
Code merged into `main` through a pull request with CI green	Pull request opened from `increment-1` ; merge after CI passes
Acceptance tests automated and passing; endpoints in the permission matrix	Done (Sections 3 and 1)
Screens in Arabic and English following the wireframes	Done
Actions write audit entries	Done (AT-AUD-01)
Design, API, or SRS updated where the implementation changed something	Done: CR-07, Design 1.3, API 1.2, 02_Design.md Section 12
Demo scenario runs without help	Done (Section 5)